



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



Kubernetes and Cloud Native Associate (KCNA) Training

TGS Ref No: TGS-2023039343

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 6.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	31 August 2023	Legacy LP: first version.	Tertiary Infotech Academy
2.0	1 October 2024	Legacy LP: second version.	Tertiary Infotech Academy
3.0	17 August 2025	Legacy LP: third version.	Tertiary Infotech Academy
4.0	17 October 2025	Legacy LP: company name updated.	Tertiary Infotech Academy
6.0	5 September 2026	Full rebuild under the Kubernetes and Cloud Native Associate (KCNA) title. Deck rebuilt to the house visual system with native editable charts and mechanism-level diagrams; 25 self-contained labs with runnable manifests, synthetic datasets and verification checks; coverage realigned to the CNCF KCNA curriculum effective 24 Nov 2025 (44/28/16/12) while preserving the registered WSQ outcomes for TSC Solution Architecture ICT-DES-4006-1.1.	Dr. Alfred Ang
6.1	10 September 2026	Visual refresh using original course-specific ImageGen cover and observability artwork; current KCNA domain blueprint, practice exam, course identity, labs and assessment mappings revalidated. No step-by-step procedures were added to the slide deck.	Dr. Alfred Ang

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Course Overview

Kubernetes and Cloud Native Associate (KCNA) Training (TGS-2023039343) is a 5-day, 37-hour instructor-led classroom course with a further 3 hours of competency assessment, 40 hours in total. It is delivered against the registered WSQ Technical Skills and Competency Solution Architecture (ICT-DES-4006-1.1) and simultaneously covers the current CNCF Kubernetes and Cloud Native Associate curriculum effective 24 November 2025 (44/28/16/12 weighting across Kubernetes Fundamentals, Container Orchestration, Cloud Native Application Delivery, Cloud Native Architecture).

Learners work on a live single-node Kubernetes cluster throughout. The course runs from a 576-slide deck (Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx) and 25 self-contained hands-on labs, each with runnable manifests, a synthetic dataset and a verification check. Class hours are 9:30 AM – 6:30 PM. Every teaching row in this plan cites the deck pages the trainer runs from, so the plan and the deck cannot drift.

Trainer: Dr. Alfred Ang, Founder & Principal Trainer, Tertiary Infotech Academy. Courseware and the assessment are hosted on the learning management system at <https://lms-tms.tertiaryinfotech.com/>.

Learning Outcomes

By the end of this course, participants will be able to:

LO1. Develop a Kubernetes architectural proof of concept.

Abilities: A1 Develop an architectural proof of concept

Knowledge: K1 Process for developing proof of concepts

LO2. Identify the technical and practical requirements in a Kubernetes setup.

Abilities: A3 Identify technical and practical requirements as well as stakeholders' demands

Knowledge: K3 Objectives of solution architecture

LO3. Develop a solution architecture within Kubernetes.

Abilities: A2 Develop a solution architecture utilising appropriate tools, techniques and models of system components and interfaces

Knowledge: K2 Components of solution architecture, K4 Steps for developing solution architecture

LO4. Prepare a technical blueprint for a Kubernetes-based solution for security and storage.

Abilities: A4 Prepare a technical blueprint for a solution in a given area

Knowledge: K6 Technical blueprint design and construction process

LO5. Demonstrate Kubernetes solution for a specific business problem.

Abilities: A5 Demonstrate how the recommended IT solutions and components collectively address an existing business problem or need

Knowledge: K5 Tools and techniques for solution architecture modelling

LO6. Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.

Abilities: A6 Implement regular system reviews to monitor solution status and make modifications, according to an architecture management framework

Knowledge: K7 Interactions among various IT components

Competency Mapping

Each registered outcome is delivered by named topics and labs and measured by a named assessment item. Slide references are read from the built deck (Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx, 576 slides).

LO	A	K	Delivered by (topics and labs)	Assessed by
LO1	A1	K1	Topic 1 Core Concepts; Module Cloud Native Architecture & the CNCF Ecosystem. Labs: Lab 01 (slide 72), Lab 02 (slide 92), Lab 03 (slide 105), Lab 04 (slide 118), Lab 05 (slide 127).	WA Q1 · PP Task 1
LO2	A3	K3	Topic 2 Workloads & Scheduling; Module Container Orchestration Fundamentals. Labs: Lab 06 (slide 155), Lab 07 (slide 175), Lab 08 (slide 190), Lab 09 (slide 211), Lab 10 (slide 237).	WA Q2 · PP Task 2
LO3	A2	K2, K4	Topic 3 Services and Networking. Labs: Lab 11 (slide 273), Lab 12 (slide 283), Lab 13 (slide 305).	WA Q3, WA Q4 · PP Task 3
LO4	A4	K6	Topic 4 Security; Topic 5 Storage. Labs: Lab 14 (slide 332), Lab 15 (slide 341), Lab 16 (slide 350), Lab 17 (slide 379), Lab 18 (slide 397), Lab 19 (slide 411).	WA Q5 · PP Task 4
LO5	A5	K5	Topic 6 Cluster Architecture, Installation & Configuration; Module Cloud Native Application Delivery. Labs: Lab 20 (slide 437), Lab 21 (slide 474).	WA Q6 · PP Task 5
LO6	A6	K7	Topic 7 Logging and Monitoring; Topic 8 Troubleshooting. Labs: Lab 22 (slide 498), Lab 23 (slide 512), Lab 24 (slide 528), Lab 25 (slide 549).	WA Q7 · PP Task 6

Written Assessment: Written Assessment (SAQ), 7 questions, 60 minutes, K-code order K1, K3, K2, K4, K6, K5, K7. Practical Performance: Practical Performance (PP), 6 tasks, 120 minutes, A-code order A1, A3, A2, A4, A5, A6.

Daily Schedule

Class hours: 9:30 AM – 6:30 PM. Every lab receives its full guide duration. Clock times govern delivery; slide references identify supporting material. Activities may continue across breaks. Breaks are excluded from the 37 training hours.

Day 1 Schedule — Cloud Native Foundations & Kubernetes Core Concepts

Deck pages 15–132 · Instructional time 8 hours

Time	Duration	Topic / Activity	Method	Slides
9:30 – 9:45 AM	15 m	Course Orientation, KCNA Blueprint and Day Map	Lecture · Demonstration · Q&A	15–20
9:45 – 10:20 AM	35 m	What 'Cloud Native' Actually Means	Lecture · Demonstration · Q&A	21–35
10:20 – 10:50 AM	30 m	What a Container Actually Is	Lecture · Demonstration · Q&A	36–48
10:50 – 11:00 AM	10 m	The API Is the Product	Lecture · Demonstration · Q&A	50–71
11:00 – 11:10 AM	10 m	<i>Morning Break</i>	<i>Break</i>	—
11:10 – 11:50 AM	40 m	The API Is the Product (continued after break)	Lecture · Demonstration · Q&A	50–71
11:50 AM – 12:35 PM	45 m	Lab 01 — Cluster Reconnaissance with kubectl · LO1/A1/K1	Hands-on Practical	72
12:35 – 1:10 PM	35 m	The Pod Is the Unit of Scheduling	Lecture · Demonstration · Q&A	74–91
1:10 – 1:50 PM	40 m	<i>Lunch Break</i>	<i>Break</i>	—
1:50 – 2:00 PM	10 m	The Pod Is the Unit of Scheduling (continued after break)	Lecture · Demonstration · Q&A	74–91
2:00 – 2:45 PM	45 m	Lab 02 — Authoring Your First Pod Manifest · LO1/A1/K1	Hands-on Practical	92
2:45 – 2:50 PM	5 m	The Pod Is the Unit of Scheduling (continued)	Lecture · Demonstration · Q&A	93–94
2:50 – 3:15 PM	25 m	Configuring a Container From Outside Its Image	Lecture · Demonstration · Q&A	95–104
3:15 – 3:50 PM	35 m	Lab 03 — Commands, Arguments and Environment · LO1/A1/K1	Hands-on Practical	105
3:50 – 4:00 PM	10 m	<i>Afternoon Break</i>	<i>Break</i>	—
4:00 – 4:05 PM	5 m	Lab 03 — Commands, Arguments and Environment · LO1/A1/K1 (continued after break)	Hands-on Practical	105
4:05 – 4:30 PM	25 m	Adding Capability Without Touching the Application	Lecture · Demonstration · Q&A	107–117
4:30 – 5:20 PM	50 m	Lab 04 — Multi-Container Pod Patterns: Sidecar, Adapter, Ambassador · LO1/A1/K1	Hands-on Practical	118
5:20 – 5:40 PM	20 m	Naming Bytes You Can Prove	Lecture · Demonstration · Q&A	119–126
5:40 – 6:20 PM	40 m	Lab 05 — Container Images, OCI and the Runtime Interface · LO1/A1/K1	Hands-on Practical	127

Time	Duration	Topic / Activity	Method	Slides
6:20 – 6:30 PM	10 m	Naming Bytes You Can Prove (continued)	Lecture · Demonstration · Q&A	128–132

Day 1 instructional total: 480 minutes = 8 hours (course_data DAY_HOURS[0] = 8 hours). Breaks excluded.

Day 2 Schedule — Workloads, Scheduling & Container Orchestration

Deck pages 133–239 · Instructional time 8 hours

Time	Duration	Topic / Activity	Method	Slides
9:30 – 10:00 AM	30 m	Container Orchestration Fundamentals	Lecture · Demonstration · Q&A	133–145
10:00 – 10:20 AM	20 m	Labels, Selectors and ReplicaSets	Lecture · Demonstration · Q&A	146–154
10:20 – 11:00 AM	40 m	Lab 06 — Labels, Selectors and ReplicaSets · LO2/A3/K3	Hands-on Practical	155
11:00 – 11:10 AM	10 m	<i>Morning Break</i>	<i>Break</i>	—
11:10 – 11:15 AM	5 m	Lab 06 — Labels, Selectors and ReplicaSets · LO2/A3/K3 (continued after break)	Hands-on Practical	155
11:15 AM – 12:00 PM	45 m	Deployments, Rollout Strategy and Rollback	Lecture · Demonstration · Q&A	157–174
12:00 – 12:55 PM	55 m	Lab 07 — Deployments, Rollout Strategy and Rollback · LO2/A3/K3	Hands-on Practical	175
12:55 – 1:10 PM	15 m	DaemonSets, Jobs, CronJobs and the Adapter Pattern	Lecture · Demonstration · Q&A	177–189
1:10 – 1:50 PM	40 m	<i>Lunch Break</i>	<i>Break</i>	—
1:50 – 2:05 PM	15 m	DaemonSets, Jobs, CronJobs and the Adapter Pattern (continued after break)	Lecture · Demonstration · Q&A	177–189
2:05 – 2:50 PM	45 m	Lab 08 — DaemonSets, Jobs and CronJobs · LO2/A3/K3	Hands-on Practical	190
2:50 – 3:35 PM	45 m	The Scheduler: Filter, Score, Bind	Lecture · Demonstration · Q&A	191–210
3:35 – 3:50 PM	15 m	Lab 09 — Scheduling: nodeSelector, Affinity, Taints and Tolerations · LO2/A3/K3	Hands-on Practical	211
3:50 – 4:00 PM	10 m	<i>Afternoon Break</i>	<i>Break</i>	—
4:00 – 4:40 PM	40 m	Lab 09 — Scheduling: nodeSelector, Affinity, Taints and Tolerations · LO2/A3/K3 (continued after break)	Hands-on Practical	211
4:40 – 5:35 PM	55 m	Resource Limits, QoS and Autoscaling	Lecture · Demonstration · Q&A	213–236
5:35 – 6:25 PM	50 m	Lab 10 — Resource Requests, Limits, QoS and Autoscaling · LO2/A3/K3	Hands-on Practical	237
6:25 – 6:30 PM	5 m	Resource Limits, QoS and Autoscaling (continued)	Lecture · Demonstration · Q&A	238–239

Day 2 instructional total: 480 minutes = 8 hours (course_data DAY_HOURS[1] = 8 hours). Breaks excluded.

Day 3 Schedule — Services, Networking & Cluster Security

Deck pages 240–359 · Instructional time 8 hours

Time	Duration	Topic / Activity	Method	Slides
9:30 – 10:20 AM	50 m	Services, Networking & Cluster Security	Lecture · Demonstration · Q&A	240–271
10:20 – 11:00 AM	40 m	Lab 11 — Services, Endpoints and Cluster DNS · LO3/A2/K2/K4	Hands-on Practical	273
11:00 – 11:10 AM	10 m	Morning Break	Break	—
11:10 – 11:20 AM	10 m	Lab 11 — Services, Endpoints and Cluster DNS · LO3/A2/K2/K4 (continued after break)	Hands-on Practical	273
11:20 – 11:35 AM	15 m	Services, Networking & Cluster Security (continued)	Lecture · Demonstration · Q&A	274–282
11:35 AM – 12:20 PM	45 m	Lab 12 — Service Types: ClusterIP, NodePort and LoadBalancer · LO3/A2/K2/K4	Hands-on Practical	283
12:20 – 12:25 PM	5 m	Services, Networking & Cluster Security (continued)	Lecture · Demonstration · Q&A	284
12:25 – 12:55 PM	30 m	Services, Networking & Cluster Security (continued)	Lecture · Demonstration · Q&A	286–304
12:55 – 1:10 PM	15 m	Lab 13 — Ingress Resources and HTTP Routing · LO3/A2/K2/K4	Hands-on Practical	305
1:10 – 1:50 PM	40 m	Lunch Break	Break	—
1:50 – 2:25 PM	35 m	Lab 13 — Ingress Resources and HTTP Routing · LO3/A2/K2/K4 (continued after break)	Hands-on Practical	305
2:25 – 2:30 PM	5 m	Services, Networking & Cluster Security (continued)	Lecture · Demonstration · Q&A	306–309
2:30 – 3:05 PM	35 m	Cluster Security	Lecture · Demonstration · Q&A	311–331
3:05 – 3:50 PM	45 m	Lab 14 — Namespaces, ServiceAccounts and RBAC · LO4/A4/K6	Hands-on Practical	332
3:50 – 4:00 PM	10 m	Afternoon Break	Break	—
4:00 – 4:10 PM	10 m	Lab 14 — Namespaces, ServiceAccounts and RBAC · LO4/A4/K6 (continued after break)	Hands-on Practical	332
4:10 – 4:25 PM	15 m	Cluster Security (continued)	Lecture · Demonstration · Q&A	333–340
4:25 – 5:15 PM	50 m	Lab 15 — Secrets, ConfigMaps and Safe Injection · LO4/A4/K6	Hands-on Practical	341
5:15 – 5:25 PM	10 m	Cluster Security (continued)	Lecture · Demonstration · Q&A	342–349
5:25 – 6:15 PM	50 m	Lab 16 — NetworkPolicy and Default-Deny Isolation · LO4/A4/K6	Hands-on Practical	350
6:15 – 6:30 PM	15 m	Cluster Security (continued)	Lecture · Demonstration · Q&A	351–359

Day 3 instructional total: 480 minutes = 8 hours (course_data DAY_HOURS[2] = 8 hours). Breaks excluded.

Day 4 Schedule — Storage, Cluster Architecture & Application Delivery

Deck pages 360–478 · Instructional time 8 hours

Time	Duration	Topic / Activity	Method	Slides
9:30 – 9:40 AM	10 m	Storage, Cluster Architecture & Application Delivery	Lecture · Demonstration · Q&A	360–363
9:40 – 10:10 AM	30 m	Storage	Lecture · Demonstration · Q&A	364–378
10:10 – 10:55 AM	45 m	Lab 17 — Volumes: emptyDir, hostPath and the Container Filesystem · LO4/A4/K6	Hands-on Practical	379
10:55 – 11:00 AM	5 m	Storage (continued)	Lecture · Demonstration · Q&A	381–396
11:00 – 11:10 AM	10 m	<i>Morning Break</i>	<i>Break</i>	—
11:10 – 11:35 AM	25 m	Storage (continued) (continued after break)	Lecture · Demonstration · Q&A	381–396
11:35 AM – 12:30 PM	55 m	Lab 18 — PersistentVolumes, Claims and StorageClasses · LO4/A4/K6	Hands-on Practical	397
12:30 – 12:55 PM	25 m	Storage (continued)	Lecture · Demonstration · Q&A	398–410
12:55 – 1:10 PM	15 m	Lab 19 — StatefulSets, Headless Services and Stable Identity · LO4/A4/K6	Hands-on Practical	411
1:10 – 1:50 PM	40 m	<i>Lunch Break</i>	<i>Break</i>	—
1:50 – 2:25 PM	35 m	Lab 19 — StatefulSets, Headless Services and Stable Identity · LO4/A4/K6 (continued after break)	Hands-on Practical	411
2:25 – 3:10 PM	45 m	Cluster Architecture, Installation & Configuration	Lecture · Demonstration · Q&A	413–436
3:10 – 3:50 PM	40 m	Lab 20 — Control Plane Anatomy and etcd Backup/Restore · LO5/A5/K5	Hands-on Practical	437
3:50 – 4:00 PM	10 m	<i>Afternoon Break</i>	<i>Break</i>	—
4:00 – 4:15 PM	15 m	Lab 20 — Control Plane Anatomy and etcd Backup/Restore · LO5/A5/K5 (continued after break)	Hands-on Practical	437
4:15 – 5:25 PM	1 h 10 m	Cloud Native Application Delivery	Lecture · Demonstration · Q&A	439–473
5:25 – 6:20 PM	55 m	Lab 21 — Packaging and Delivery: Helm, Kustomize and GitOps · LO5/A5/K5	Hands-on Practical	474
6:20 – 6:30 PM	10 m	Cloud Native Application Delivery (continued)	Lecture · Demonstration · Q&A	475–478

Day 4 instructional total: 480 minutes = 8 hours (course_data DAY_HOURS[3] = 8 hours). Breaks excluded.

Day 5 Schedule — Observability, Troubleshooting & Assessment

Deck pages 479–568 · Instructional time 5 hours · Assessment 3 hours

Time	Duration	Topic / Activity	Method	Slides
9:30 – 9:55 AM	25 m	Logging and Monitoring	Lecture · Demonstration · Q&A	479–497
9:55 – 10:40 AM	45 m	Lab 22 — Probes, Health and Self-Healing · LO6/A6/K7	Hands-on Practical	498
10:40 – 10:55 AM	15 m	Logging and Monitoring (continued)	Lecture · Demonstration · Q&A	500–511
10:55 – 11:00 AM	5 m	Lab 23 — Events, Logs, Field Selectors and the Metrics Server · LO6/A6/K7	Hands-on Practical	512
11:00 – 11:10 AM	10 m	<i>Morning Break</i>	<i>Break</i>	—
11:10 – 11:55 AM	45 m	Lab 23 — Events, Logs, Field Selectors and the Metrics Server · LO6/A6/K7 (continued after break)	Hands-on Practical	512
11:55 AM – 12:15 PM	20 m	Logging and Monitoring (continued)	Lecture · Demonstration · Q&A	513–527
12:15 – 1:00 PM	45 m	Lab 24 — Metrics, Prometheus Exposition and the Observability Pipeline · LO6/A6/K7	Hands-on Practical	528
1:00 – 1:10 PM	10 m	Troubleshooting	Lecture · Demonstration · Q&A	529–548
1:10 – 1:50 PM	40 m	<i>Lunch Break</i>	<i>Break</i>	—
1:50 – 2:05 PM	15 m	Troubleshooting (continued after break)	Lecture · Demonstration · Q&A	529–548
2:05 – 3:00 PM	55 m	Lab 25 — Troubleshooting Triage: Application, Node and Control Plane · LO6/A6/K7	Hands-on Practical	549
3:00 – 3:10 PM	10 m	Ecosystem, Principles and Community	Lecture · Demonstration · Q&A	551–558
3:10 – 3:20 PM	10 m	Course Recap and Exam Readiness	Lecture · Demonstration · Q&A	559–567
3:20 – 3:30 PM	10 m	<i>Afternoon Break</i>	<i>Break</i>	—
3:30 – 4:30 PM	1 h 00 m	<i>Assessment 1 — Written Assessment (SAQ)</i>	<i>Assessment</i>	—
4:30 – 6:30 PM	2 h 00 m	<i>Assessment 2 — Practical Performance (PP)</i>	<i>Assessment</i>	—

Day 5 instructional total: 300 minutes = 5 hours (course_data DAY_HOURS[4] = 5 hours). Breaks excluded; assessment 3 hours is additional.

Assessment block on Day 5 (shown in the table above): 3:30 – 4:30 PM Assessment 1 — Written Assessment (SAQ); 4:30 – 6:30 PM Assessment 2 — Practical Performance (PP).

Topic-by-Topic Breakdown — Day 1: Cloud Native Foundations & Kubernetes Core Concepts

Deck pages 15–132 · 118 slides · 5 labs · 8 instructional hours

Course Orientation, KCNA Blueprint and Day Map · Slides 15–20

Registered Topic 1: Core Concepts · LO1 / A1 / K1 · 6 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

What 'Cloud Native' Actually Means · Slides 21–35

KCNA module: Cloud Native Architecture & the CNCF Ecosystem · LO1 / A1 / K1 · 15 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

What a Container Actually Is · Slides 36–48

KCNA module: Cloud Native Architecture & the CNCF Ecosystem · LO1 / A1 / K1 · 13 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

The API Is the Product · Slides 50–72

Registered Topic 1: Core Concepts · LO1 / A1 / K1 · 23 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 01 — Cluster Reconnaissance with kubectl · Slide 72

Item	Detail
Deck slide	Slide 72 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab01
LO / A / K	LO1 / A1 / K1
Outcome	Develop a Kubernetes architectural proof of concept.
Repository path	courseware/labs/lab-01-kubectl-and-cluster-recon/
Learner README	courseware/labs/lab-01-kubectl-and-cluster-recon/README.md
Assessment link	WA Q1 · PP Task 1

The Pod Is the Unit of Scheduling · Slides 74–94

Registered Topic 1: Core Concepts · LO1 / A1 / K1 · 21 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 02 — Authoring Your First Pod Manifest · Slide 92

Item	Detail
Deck slide	Slide 92 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab02
LO / A / K	LO1 / A1 / K1
Outcome	Develop a Kubernetes architectural proof of concept.
Repository path	courseware/labs/lab-02-first-pod-manifest/
Learner README	courseware/labs/lab-02-first-pod-manifest/README.md
Assessment link	WA Q1 · PP Task 1

Configuring a Container From Outside Its Image · Slides 95–105

Registered Topic 1: Core Concepts · LO1 / A1 / K1 · 11 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 03 — Commands, Arguments and Environment · Slide 105

Item	Detail
Deck slide	Slide 105 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	40 minutes
Namespace	kcna-lab03
LO / A / K	LO1 / A1 / K1
Outcome	Develop a Kubernetes architectural proof of concept.
Repository path	courseware/labs/lab-03-commands-args-env/
Learner README	courseware/labs/lab-03-commands-args-env/README.md
Assessment link	WA Q1 · PP Task 1

Adding Capability Without Touching the Application · Slides 107–118

Registered Topic 1: Core Concepts · LO1 / A1 / K1 · 12 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 04 — Multi-Container Pod Patterns: Sidecar, Adapter, Ambassador · Slide 118

Item	Detail
Deck slide	Slide 118 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab04
LO / A / K	LO1 / A1 / K1
Outcome	Develop a Kubernetes architectural proof of concept.
Repository path	courseware/labs/lab-04-multi-container-patterns/
Learner README	courseware/labs/lab-04-multi-container-patterns/README.md
Assessment link	WA Q1 · PP Task 1

Naming Bytes You Can Prove · Slides 119–132

KCNA module: Cloud Native Architecture & the CNCF Ecosystem · LO1 / A1 / K1 · 14 slides

Develop a Kubernetes architectural proof of concept.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 05 — Container Images, OCI and the Runtime Interface · Slide 127

Item	Detail
Deck slide	Slide 127 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	40 minutes
Namespace	kcna-lab05
LO / A / K	LO1 / A1 / K1
Outcome	Develop a Kubernetes architectural proof of concept.
Repository path	courseware/labs/lab-05-container-images-oci/
Learner README	courseware/labs/lab-05-container-images-oci/README.md
Assessment link	WA Q1 · PP Task 1

Topic-by-Topic Breakdown — Day 2: Workloads, Scheduling & Container Orchestration

Deck pages 133–239 · 107 slides · 5 labs · 8 instructional hours

Container Orchestration Fundamentals · Slides 133–145

KCNA module: Container Orchestration Fundamentals · LO2 / A3 / K3 · 13 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

Labels, Selectors and ReplicaSets · Slides 146–155

Registered Topic 2: Workloads & Scheduling · LO2 / A3 / K3 · 10 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 06 — Labels, Selectors and ReplicaSets · Slide 155

Item	Detail
Deck slide	Slide 155 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab06
LO / A / K	LO2 / A3 / K3
Outcome	Identify the technical and practical requirements in a Kubernetes setup.
Repository path	courseware/labs/lab-06-labels-selectors-replicasets/
Learner README	courseware/labs/lab-06-labels-selectors-replicasets/README.md
Assessment link	WA Q2 · PP Task 2

Deployments, Rollout Strategy and Rollback · Slides 157–175

Registered Topic 2: Workloads & Scheduling · LO2 / A3 / K3 · 19 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 07 — Deployments, Rollout Strategy and Rollback · Slide 175

Item	Detail
Deck slide	Slide 175 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab07
LO / A / K	LO2 / A3 / K3
Outcome	Identify the technical and practical requirements in a Kubernetes setup.
Repository path	courseware/labs/lab-07-deployments-rollout-rollback/
Learner README	courseware/labs/lab-07-deployments-rollout-rollback/README.md
Assessment link	WA Q2 · PP Task 2

DaemonSets, Jobs, CronJobs and the Adapter Pattern · Slides 177–190

Registered Topic 2: Workloads & Scheduling · LO2 / A3 / K3 · 14 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 08 — DaemonSets, Jobs and CronJobs · Slide 190

Item	Detail
Deck slide	Slide 190 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab08
LO / A / K	LO2 / A3 / K3
Outcome	Identify the technical and practical requirements in a Kubernetes setup.
Repository path	courseware/labs/lab-08-daemonsets-jobs-cronjobs/
Learner README	courseware/labs/lab-08-daemonsets-jobs-cronjobs/README.md
Assessment link	WA Q2 · PP Task 2

The Scheduler: Filter, Score, Bind · Slides 191–211

Registered Topic 2: Workloads & Scheduling · LO2 / A3 / K3 · 21 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 09 — Scheduling: nodeSelector, Affinity, Taints and Tolerations · Slide 211

Item	Detail
Deck slide	Slide 211 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab09
LO / A / K	LO2 / A3 / K3
Outcome	Identify the technical and practical requirements in a Kubernetes setup.
Repository path	courseware/labs/lab-09-scheduling-affinity-taints/
Learner README	courseware/labs/lab-09-scheduling-affinity-taints/README.md
Assessment link	WA Q2 · PP Task 2

Resource Limits, QoS and Autoscaling · Slides 213–239

KCNA module: Container Orchestration Fundamentals · LO2 / A3 / K3 · 27 slides

Identify the technical and practical requirements in a Kubernetes setup.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 10 — Resource Requests, Limits, QoS and Autoscaling · Slide 237

Item	Detail
Deck slide	Slide 237 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab10
LO / A / K	LO2 / A3 / K3
Outcome	Identify the technical and practical requirements in a Kubernetes setup.
Repository path	courseware/labs/lab-10-resources-requests-limits-autoscaling/
Learner README	courseware/labs/lab-10-resources-requests-limits-autoscaling/README.md
Assessment link	WA Q2 · PP Task 2

Topic-by-Topic Breakdown — Day 3: Services, Networking & Cluster Security

Deck pages 240–359 · 120 slides · 6 labs · 8 instructional hours

Services, Networking & Cluster Security · Slides 240–309

Registered Topic 3: Services and Networking · LO3 / A2 / K2, K4 · 70 slides

Develop a solution architecture within Kubernetes.

Delivery: Lecture · Demonstration · Q&A; 3 hands-on labs in this module.

Lab 11 — Services, Endpoints and Cluster DNS · Slide 273

Item	Detail
Deck slide	Slide 273 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab11
LO / A / K	LO3 / A2 / K2, K4
Outcome	Develop a solution architecture within Kubernetes.
Repository path	courseware/labs/lab-11-services-endpoints/
Learner README	courseware/labs/lab-11-services-endpoints/README.md
Assessment link	WA Q3, WA Q4 · PP Task 3

Lab 12 — Service Types: ClusterIP, NodePort and LoadBalancer · Slide 283

Item	Detail
Deck slide	Slide 283 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab12
LO / A / K	LO3 / A2 / K2, K4
Outcome	Develop a solution architecture within Kubernetes.
Repository path	courseware/labs/lab-12-service-types-nodeport-lb/
Learner README	courseware/labs/lab-12-service-types-nodeport-lb/README.md
Assessment link	WA Q3, WA Q4 · PP Task 3

Lab 13 — Ingress Resources and HTTP Routing · Slide 305

Item	Detail
Deck slide	Slide 305 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab13
LO / A / K	LO3 / A2 / K2, K4
Outcome	Develop a solution architecture within Kubernetes.
Repository path	courseware/labs/lab-13-ingress-routing/
Learner README	courseware/labs/lab-13-ingress-routing/README.md
Assessment link	WA Q3, WA Q4 · PP Task 3

Cluster Security · Slides 311–359

Registered Topic 4: Security · LO4 / A4 / K6 · 49 slides

Prepare a technical blueprint for a Kubernetes-based solution for security and storage.

Delivery: Lecture · Demonstration · Q&A; 3 hands-on labs in this module.

Lab 14 — Namespaces, ServiceAccounts and RBAC · Slide 332

Item	Detail
Deck slide	Slide 332 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab14
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-14-namespaces-serviceaccounts-rbac/
Learner README	courseware/labs/lab-14-namespaces-serviceaccounts-rbac/README.md
Assessment link	WA Q5 · PP Task 4

Lab 15 — Secrets, ConfigMaps and Safe Injection · Slide 341

Item	Detail
Deck slide	Slide 341 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab15
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-15-secrets-configmaps/
Learner README	courseware/labs/lab-15-secrets-configmaps/README.md
Assessment link	WA Q5 · PP Task 4

Lab 16 — NetworkPolicy and Default-Deny Isolation · Slide 350

Item	Detail
Deck slide	Slide 350 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab16
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-16-networkpolicy-isolation/
Learner README	courseware/labs/lab-16-networkpolicy-isolation/README.md
Assessment link	WA Q5 · PP Task 4

Topic-by-Topic Breakdown — Day 4: Storage, Cluster Architecture & Application Delivery

Deck pages 360–478 · 119 slides · 5 labs · 8 instructional hours

Storage, Cluster Architecture & Application Delivery · Slides 360–363

Registered Topic 5: Storage · LO4 / A4 / K6 · 4 slides

Prepare a technical blueprint for a Kubernetes-based solution for security and storage.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

Storage · Slides 364–411

Registered Topic 5: Storage · LO4 / A4 / K6 · 48 slides

Prepare a technical blueprint for a Kubernetes-based solution for security and storage.

Delivery: Lecture · Demonstration · Q&A; 3 hands-on labs in this module.

Lab 17 — Volumes: emptyDir, hostPath and the Container Filesystem · Slide 379

Item	Detail
Deck slide	Slide 379 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab17
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-17-volumes-emptydir-hostpath/
Learner README	courseware/labs/lab-17-volumes-emptydir-hostpath/README.md
Assessment link	WA Q5 · PP Task 4

Lab 18 — PersistentVolumes, Claims and StorageClasses · Slide 397

Item	Detail
Deck slide	Slide 397 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab18
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-18-pv-pvc-storageclass/
Learner README	courseware/labs/lab-18-pv-pvc-storageclass/README.md
Assessment link	WA Q5 · PP Task 4

Lab 19 — StatefulSets, Headless Services and Stable Identity · Slide 411

Item	Detail
Deck slide	Slide 411 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab19
LO / A / K	LO4 / A4 / K6
Outcome	Prepare a technical blueprint for a Kubernetes-based solution for security and storage.
Repository path	courseware/labs/lab-19-statefulset-headless/
Learner README	courseware/labs/lab-19-statefulset-headless/README.md
Assessment link	WA Q5 · PP Task 4

Cluster Architecture, Installation & Configuration · Slides 413–437

Registered Topic 6: Cluster Architecture, Installation & Configuration · LO5 / A5 / K5 · 25 slides

Demonstrate Kubernetes solution for a specific business problem.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 20 — Control Plane Anatomy and etcd Backup/Restore · Slide 437

Item	Detail
Deck slide	Slide 437 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab20
LO / A / K	LO5 / A5 / K5
Outcome	Demonstrate Kubernetes solution for a specific business problem.
Repository path	courseware/labs/lab-20-cluster-architecture-etcd/
Learner README	courseware/labs/lab-20-cluster-architecture-etcd/README.md
Assessment link	WA Q6 · PP Task 5

Cloud Native Application Delivery · Slides 439–478

KCNA module: Cloud Native Application Delivery · LO5 / A5 / K5 · 40 slides

Demonstrate Kubernetes solution for a specific business problem.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 21 — Packaging and Delivery: Helm, Kustomize and GitOps · Slide 474

Item	Detail
Deck slide	Slide 474 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab21
LO / A / K	LO5 / A5 / K5
Outcome	Demonstrate Kubernetes solution for a specific business problem.
Repository path	courseware/labs/lab-21-helm-kustomize-gitops/
Learner README	courseware/labs/lab-21-helm-kustomize-gitops/README.md
Assessment link	WA Q6 · PP Task 5

Topic-by-Topic Breakdown — Day 5: Observability, Troubleshooting & Assessment

Deck pages 479–568 · 90 slides · 4 labs · 5 instructional hours

Logging and Monitoring · Slides 479–528

Registered Topic 7: Logging and Monitoring · LO6 / A6 / K7 · 50 slides

Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.

Delivery: Lecture · Demonstration · Q&A; 3 hands-on labs in this module.

Lab 22 — Probes, Health and Self-Healing · Slide 498

Item	Detail
Deck slide	Slide 498 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab22
LO / A / K	LO6 / A6 / K7
Outcome	Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.
Repository path	courseware/labs/lab-22-probes-health-selfhealing/
Learner README	courseware/labs/lab-22-probes-health-selfhealing/README.md
Assessment link	WA Q7 · PP Task 6

Lab 23 — Events, Logs, Field Selectors and the Metrics Server · Slide 512

Item	Detail
Deck slide	Slide 512 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	50 minutes
Namespace	kcna-lab23
LO / A / K	LO6 / A6 / K7
Outcome	Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.
Repository path	courseware/labs/lab-23-events-logs-metrics/
Learner README	courseware/labs/lab-23-events-logs-metrics/README.md
Assessment link	WA Q7 · PP Task 6

Lab 24 — Metrics, Prometheus Exposition and the Observability Pipeline · Slide 528

Item	Detail
Deck slide	Slide 528 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	45 minutes
Namespace	kcna-lab24
LO / A / K	LO6 / A6 / K7
Outcome	Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.
Repository path	courseware/labs/lab-24-prometheus-observability/
Learner README	courseware/labs/lab-24-prometheus-observability/README.md
Assessment link	WA Q7 · PP Task 6

Troubleshooting · Slides 529–549

Registered Topic 8: Troubleshooting · LO6 / A6 / K7 · 21 slides

Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.

Delivery: Lecture · Demonstration · Q&A; 1 hands-on lab in this module.

Lab 25 — Troubleshooting Triage: Application, Node and Control Plane · Slide 549

Item	Detail
Deck slide	Slide 549 (lab card in Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx)
Guide duration	55 minutes
Namespace	kcna-lab25
LO / A / K	LO6 / A6 / K7
Outcome	Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.
Repository path	courseware/labs/lab-25-troubleshooting-triage/
Learner README	courseware/labs/lab-25-troubleshooting-triage/README.md
Assessment link	WA Q7 · PP Task 6

Ecosystem, Principles and Community · Slides 551–558

Registered Topic 7: Logging and Monitoring · LO6 / A6 / K7 · 8 slides

Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

Course Recap and Exam Readiness · Slides 559–567

Registered Topic 8: Troubleshooting · LO6 / A6 / K7 · 9 slides

Implement regular monitoring of the Kubernetes system and perform necessary troubleshooting.

Delivery: Lecture · Demonstration · Q&A; no lab in this module.

Resources Required

Trainer Requirements

- Practising Kubernetes engineer at Founder & Principal Trainer, Tertiary Infotech Academy level, holding a current ACTA/ACLP certification, with demonstrated cloud native delivery experience. Course author and lead trainer: Dr. Alfred Ang.
- Presentation deck Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx (576 slides) and this Lesson Plan, both at version v6.1.
- Laptop with a working single-node cluster, a projector or shared screen at 1920x1080 or better, and a terminal at a font size readable from the back of the room.
- Lab repository cloned locally: courseware/labs/ (all 25 labs, each with manifests, dataset and verifier).
- Trainer access to the LMS at <https://lms-tms.tertiaryinfotech.com/> to release courseware and open the assessment on Day 5.
- A device for course-run attendance capture and the separate TRAQOM course-quality feedback survey.

Learner Requirements

- Laptop, 8 GB RAM minimum (16 GB recommended), 20 GB free disk, administrator rights to install container tooling.
- Stable internet connection for image pulls and LMS access.
- Comfort with a command line and a text editor; no prior Kubernetes experience is assumed.
- LMS account at <https://lms-tms.tertiaryinfotech.com/> for courseware download and assessment submission.
- Optional self-study: timed KCNA practice questions at <https://exams.tertiaryinfotech.com/practice-exams/linuxfoundation/linuxfoundation-kcna>.

Cluster Setup

- One of: kind (Kubernetes in Docker), minikube, or Docker Desktop with Kubernetes enabled. kind is the reference environment used to author and verify the labs.
- Docker Engine or Docker Desktop as the container runtime host.
- kubectl matching the cluster minor version, on PATH.
- Helm and Kustomize (kubectl -k) for the Day 4 delivery lab.
- metrics-server for the Day 5 observability labs; a CNI that enforces NetworkPolicy for the Day 3 isolation lab.
- Each lab runs in its own namespace (kcna-lab01 through kcna-lab25), so labs never collide and any lab can be reset independently.

Lab Repository

- Path in this courseware bundle: courseware/labs/
- Public repository: <https://github.com/tertiarycourses/TGS-2023039343-Kubernetes-and-Cloud-Native-Associate-KCNA-Training>
- 25 labs, one folder per lab, each containing a README, runnable manifests, a synthetic dataset and a verification check.

- Learners clone once on Day 1 and use the same working copy for the whole course and for the Practical Performance assessment.

Learning Management System

- LMS: <https://lms-tms.tertiaryinfotech.com/>
- Hosts the slide deck, the Learner Guide, the lab links and both assessment instruments.
- Course page: <https://www.tertiarycourses.com.sg/wsq-kubernetes-and-cloud-native-associate-kcna-training.html>
- Enquiries: enquiry@tertiaryinfotech.com · +65 6100 0613

Assessment

Assessment is competency-based and graded Competent (C) or Not Yet Competent (NYC). Both instruments are open book. The total assessment duration is 3 hours (180 minutes), conducted on the afternoon of Day 5 after the recap and briefing session. Learners must attain C in every outcome to be assessed Competent overall; a learner assessed NYC in any item is offered one re-assessment of that item.

Instruments

Instrument	Duration	Items	Scheduled	Coverage
Written Assessment (SAQ)	60 minutes	7 open-ended questions	3:30 – 4:30 PM	Knowledge codes in order: K1, K3, K2, K4, K6, K5, K7
Practical Performance (PP)	120 minutes	6 practical tasks	4:30 – 6:30 PM	Outcomes in order: LO1, LO2, LO3, LO4, LO5, LO6; abilities A1, A3, A2, A4, A5, A6

Recap, Briefing and Assessment Schedule on Day 5

- 3:30 – 4:30 PM · 1 h 00 m · Assessment 1 — Written Assessment (SAQ)
- 4:30 – 6:30 PM · 2 h 00 m · Assessment 2 — Practical Performance (PP)

These times are generated from `course_data.day_schedule(5)`; the class window is 9:30 AM – 6:30 PM.

Written Assessment (SAQ)

- 7 short-answer, open-ended questions covering the seven knowledge codes K1 to K7 in the order K1, K3, K2, K4, K6, K5, K7.
- 60 minutes, open book. Learners may refer to the slide deck, the Learner Guide, the lab READMEs and the official Kubernetes documentation.
- Answers are typed into the LMS assessment form and submitted before the timer closes.
- Marked against the model answer key; each question is graded C or NYC.

Practical Performance (PP)

- 6 practical tasks, one per learning outcome (LO1, LO2, LO3, LO4, LO5, LO6), exercising abilities A1, A3, A2, A4, A5, A6.
- 120 minutes, open book, performed on the learner's own cluster using the same lab repository used across the five days.
- Each task is verified by observable cluster state, not by narrative; the assessor confirms the objects and behaviour the task requires.
- Evidence (manifests, command transcript, kubectl output) is uploaded to the LMS.

Assessment Flow

- Step 1. Course and assessment attendance records are captured through the designated process. TRAQOM course-quality feedback is collected separately.
- Step 2. Trainer confirms the attendance record meets the minimum 75% attendance requirement for funding eligibility.
- Step 3. Assessment briefing during Session 3 on Day 5 — instruments, timings, open-book rules and the grading basis.
- Step 4. Written Assessment (SAQ) opens on the LMS for 60 minutes.
- Step 5. Practical Performance (PP) opens for 120 minutes; learners work on their own cluster.
- Step 6. Learners submit answers and evidence on the LMS.

- Step 7. Learners sign the assessment record; the assessor records C / NYC per item and overall.
- Step 8. Any NYC outcome is addressed through the provider's reassessment process.

Funding and Eligibility Criteria

- Minimum 75% attendance across the course.
- Competent (C) in all assessment items.
- Assessment and courseware are delivered through the LMS at <https://lms-tms.tertiaryinfotech.com/>.
- Course reference: TGS-2023039343 · TSC Solution Architecture ICT-DES-4006-1.1.

Deck cross-reference: this Lesson Plan cites Kubernetes and Cloud Native Associate (KCNA) Training-v6.1.pptx (576 slides); all 25 lab slide numbers are taken from qa/slide_map.json emitted by the deck build.